

Testing Wi-Fi (802.11n/ac) Against Fast Ethernet Under Wall Attenuation and RF Interference

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Project Artifacts: <https://github.com/arthurkrasilnikov/wifi-vs-ethernet-performance-benchmark>

Abstract—Theoretical Wi-Fi link rates advertised by networking hardware manufacturers rarely reflect achievable transport-layer throughput in typical indoor environments. In this project, I measured TCP throughput, UDP bandwidth, packet loss, and latency jitter across 2.4 GHz (802.11n), 5.0 GHz (802.11ac), and 100 Mbps Fast Ethernet baselines using iperf3 on Windows. Tests were executed across line-of-sight positioning, progressive structural wall attenuation, and intentional non-Wi-Fi ISM band interference. The empirical data confirms that while 5.0 GHz matches wired line rates at close range, its throughput drops by over 50% through interior obstacles. Furthermore, 2.4 GHz channels exhibit lower overall bandwidth and severe jitter degradation under continuous RF contention.

Index Terms—802.11ac, 802.11n, Fast Ethernet, iperf3, Latency Jitter, Signal Attenuation, Network Benchmark.

wlan show interfaces prior to each run. Signal quality values reported by the driver were converted to received power estimates in dBm:

$$RSSI(dBm) = \frac{Signal(\%)}{2} - 100 \quad (1)$$

Network saturation was measured using standard iperf3 socket streams. Peak transport throughput was evaluated with 15-second single-stream TCP transfers. To measure latency stability and frame integrity under load, 20-second UDP streams were injected at 50 Mbps and 100 Mbps target rates, recording packet drop rates and jitter over three separate runs for each location. All baseline and wireless metrics collected across the scenarios are summarized in Table I.

I. INTRODUCTION

Consumer wireless deployments frequently encounter noticeable performance mismatches between advertised PHY link rates and upper-layer socket throughput. Many entry-to-mid-tier dual-band access points pair high-bandwidth 802.11ac radios with legacy 100 Mbps physical Ethernet switches.

The goal of this project was to establish an empirical benchmark isolating how physical switching limits, structural path loss, and carrier-sense contention affect actual application performance compared to a reference wired link.

II. HARDWARE AND TESTBED SETUP

All benchmarks were performed within a controlled residential layout using consumer networking gear. The testbed relies on a TP-Link Archer C20 v5.0 access point featuring four 100BASE-TX Fast Ethernet switch ports. The client station is equipped with a Realtek 8821AE PCIe wireless adapter operating in a single-stream 1×1 SISO configuration.

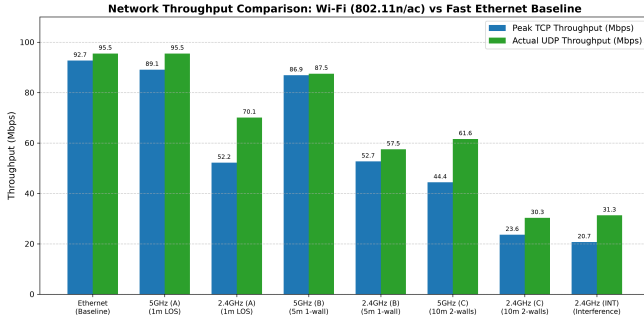
For the 5.0 GHz tests, the router was configured on Channel 44 (UNII-1) with an 80 MHz channel width, yielding a negotiated PHY link rate of 433.3 Mbps under 802.11ac. The 2.4 GHz network was set to Channel 10 with a dynamic 20/40 MHz width under 802.11n, negotiating a maximum PHY rate of 150.0 Mbps. A direct Cat 5e patch cable operating at 100BASE-TX Full Duplex served as the wired control reference.

III. MEASUREMENT METHODOLOGY

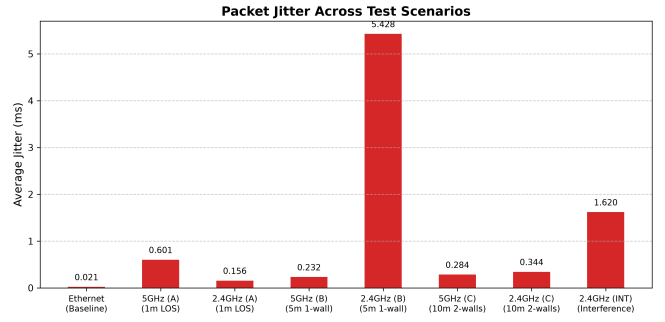
To keep measurement intervals consistent and eliminate operator latency, benchmarking was automated using Windows batch scripts. Link parameters were recorded via `netsh`

TABLE I: Measured Transport Layer Performance Dataset

Test ID	Band / Type	RSSI (dBm)	TCP (Mbps)	UDP (Mbps)	Loss (%)	Jitter (ms)
ETH_01	Wired Cat 5e	0	92.70	95.50	0.02	0.021
W50_A	5 GHz (1m LOS)	-50	89.10	95.50	4.31	0.601
W24_A	2.4 GHz (1m LOS)	-50	52.20	70.10	8.38	0.156
W50_B	5 GHz (5m, 1 Wall)	-50	86.90	87.47	11.98	0.232
W24_B	2.4 GHz (5m, 1 Wall)	-50	52.70	57.50	0.01	5.428
W50_C	5 GHz (10m, 2 Walls)	-68	44.40	61.63	0.32	0.284
W24_C	2.4 GHz (10m, 2 Walls)	-50	23.60	30.33	0.01	0.344
W24_INT	2.4 GHz (+ ISM Noise)	-51.5	20.70	31.33	0.00	1.620



(a) Peak TCP and UDP throughput across conditions.



(b) Average UDP latency jitter profile across conditions.

Fig. 1: Empirical transport layer benchmarking data across wireless and wired testbeds.

IV. TECHNICAL ANALYSIS AND DISCUSSION

A. Physical Layer Switching Bottleneck

In scenario W50_A with 1 m direct line-of-sight, the 5.0 GHz 802.11ac link achieved 89.10 Mbps TCP throughput and 95.50 Mbps UDP throughput, virtually mirroring the direct wired Cat 5e baseline (ETH_01).

Although the client wireless adapter established a physical layer (PHY) rate of 433.3 Mbps over an 80 MHz wide channel, application throughput reached an immutable ceiling. This behavior occurs because the access point relies on an internal 100BASE-TX Fast Ethernet MII switch fabric. Data traveling between the wired server host and the wireless bridge is restricted by the physical switch port's 100 Mbps line rate, proving that wireless improvements provide zero benefit when bounded by upstream wired interfaces.

B. Frequency Attenuation and Penetration

Structural walls introduced clear frequency-dependent trade-offs across the testing points. In scenario W50_C at a 10 m distance through two interior walls, the 5.0 GHz RSSI dropped from -50 dBm down to -68 dBm. To maintain connection stability, the Realtek adapter dynamically downshifted its Modulation and Coding Scheme (MCS), decreasing TCP throughput by 50.1% down to 44.40 Mbps.

Conversely, the 2.4 GHz carrier retained stable signal reception (-50 dBm) at Point C due to its longer wavelength. However, narrow channel constraints and baseline protocol overhead capped TCP throughput at 23.60 Mbps.

C. Interference and Channel Contention

During test scenario W24_INT, an active non-Wi-Fi electromagnetic source operating in the 2.4 GHz ISM band was introduced at Point C.

While the packet drop rate remained at 0.00%, the average UDP jitter increased by 470% from 0.344 ms to 1.620 ms, with transient spikes reaching 3.89 ms. Under the IEEE 802.11 CSMA/CA MAC mechanism, RF energy in the channel forces clear channel assessment (CCA) deferrals and exponential random backoff timers. This media contention reduced TCP throughput from 23.60 Mbps to 20.70 Mbps, showing how latency variance degrades connection stability before actual packet loss occurs.

V. SUMMARY AND CONCLUSION

The empirical data demonstrates distinct operational regimes across the physical media. Wired Fast Ethernet maintains a near-zero jitter floor (0.021 ms) with negligible loss, remaining essential for latency-critical applications. In contrast, 5.0 GHz 802.11ac delivers line-rate throughput in proximity but suffers sharp degradation through dense structural obstacles. Finally, 2.4 GHz 802.11n preserves base coverage at the expense of acute susceptibility to RF contention.

Real-world network performance is primarily dictated by intermediate switching hardware, RF attenuation, and media contention rather than maximum advertised PHY rates. Raw logs, batch automation scripts, and reproducible visualization routines are published at: <https://github.com/arthurkrasilnikov/wifi-vs-ethernet-performance-benchmark>.